

Geometric Sequences and Series (7.3)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 4

_____ Find the common ratio of 2, 6, 18, 54, ...

ANSWER: $\frac{1}{2}$

_____ Find S_3 of 8, 4, 2.

ANSWER: $3 \cdot 2^{n-1}$

_____ Find a_5 for $a_1 = 2$, $r = 3$.

ANSWER: no

_____ Find a_4 for $a_1 = 1$, $r = -2$.

ANSWER: 3

_____ Write an explicit formula for the geometric sequence with $a_1 = 3$, $r = 2$.

ANSWER: $\frac{9}{2}$

_____ Does the infinite series $\sum_{n=1}^{\infty} 2^n$ converge?

ANSWER: 2

_____ Find the common ratio of 16, 8, 4, 2, ...

ANSWER: -8

_____ Find $\sum_{n=1}^{\infty} 4\left(\frac{1}{2}\right)^n$.

ANSWER: 162

_____ Find the sum $1 + \frac{1}{2} + \frac{1}{4} + \cdots$

ANSWER: 14

_____ Find the sum $3 + 1 + \frac{1}{3} + \cdots$